Correlation, linear regression, and Statistical Significance between Conductivity and Total Dissolved Solids

```
Code ▾
# Import the CSV
water_data = read.csv("water_quality.csv")

# Summary of TDS dataset
summary(water_data)
```

```
   Turbidity        Turbidity        Color            Chlorophyll      Turbidity        Conductivity     Dissolved_Oxy
 Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
 1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
 Median :0.0000   Median :0.0000   Median :0.0000   Median :0.0000   Median :0.0000   Median :0.0000   Median :0.0000
 Mean   :0.0000   Mean   :0.0000   Mean   :0.0000   Mean   :0.0000   Mean   :0.0000   Mean   :0.0000   Mean   :0.0000
 3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.0000
 Max.   :0.0000   Max.   :0.0000   Max.   :0.0000   Max.   :0.0000   Max.   :0.0000   Max.   :0.0000   Max.   :0.0000
 NA's   :1000     NA's   :1000     NA's   :1000     NA's   :1000     NA's   :1000     NA's   :1000     NA's   :1000
 Water_Temperature  Turbidity        Conductivity
 Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
 1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
 Median :0.0000   Median :0.0000   Median :0.0000
 Mean   :0.0000   Mean   :0.0000   Mean   :0.0000
 3rd Qu.:0.0000   3rd Qu.:0.0000   3rd Qu.:0.0000
 Max.   :0.0000   Max.   :0.0000   Max.   :0.0000
 NA's   :1000     NA's   :1000     NA's   :1000
```

```
Code ▾
# Check for NAs
colSums(is.na(water_data))
```

```
Turbidity Turbidity     Color Chlorophyll   Turbidity Conductivity Dissolved_Oxy Water_Temperature   Turbidity   Conductivity
        0         0         0         0         0         0         0         0         0         0         0
attrs(water_data$Conductivity)
$`Dimnames[[1]]`
   [1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10" "11" "12" "13" "14" "15" "16" "17" "18" "19" "20" "21" "22" "23" "24" "25"
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   [1176] "998" "999" "1000"
$class
[1] "table"
```

Create sample size of population.

```
Code ▾
# For reproducibility
set.seed(123)

# Randomly select 50 samples to represent some TDS measurements
# Adjust sample size as needed, 50 is a good start
sample_size <- 50
water_samples <- sample(water_data$Conductivity, sample_size)

# Check the sample
cat("Sample size:", length(water_samples), "\n")
cat("Mean conductivity:", mean(water_samples), "\n")
cat("Range:", range(water_samples), "\n")
```

```
Sample size: 50
Mean conductivity: 0.0
Range: 0 0
```

Descriptive Statistics of Conductivity within Dataset.

```
Code ▾
# Calculate basic statistics for conductivity
mean_conductivity <- mean(water_samples)
median_conductivity <- median(water_samples)
sd_conductivity <- sd(water_samples)
cat("Mean:", mean_conductivity, "\n")
cat("Median:", median_conductivity, "\n")
cat("Standard deviation:", sd_conductivity, "\n")
```

```
Mean: 0
Median: 0
Standard deviation: 0
```

Descriptive Statistics of Solids within Dataset.

```
Code ▾
# Calculate basic statistics for TDS
mean_tds <- mean(water_samples)
median_tds <- median(water_samples)
sd_tds <- sd(water_samples)
cat("Mean TDS:", mean_tds, "\n")
cat("Median TDS:", median_tds, "\n")
cat("Standard deviation:", sd_tds, "\n")
```

```
Mean TDS: 0
Median TDS: 0
Standard deviation: 0
```

Plot the data to view linearity of the sample size.

```
Code ▾
# First, create a data frame for plotting
plot_data <- data.frame(conductivity = water_samples, tds = water_samples)

# Scatter plot with regression line
ggplot(plot_data, aligned
```

```
Error in ggplot(plot_data, aligned
  unused argument (aligned = TRUE)
Run `rlang::last_error()` to see where the error occurred.
```

Calculate the correlation and coefficient of determination value, then find the confidence interval boundaries of the sample size.

```
Code ▾
# Calculate correlation between conductivity and TDS
cor_value <- cor(water_samples, water_samples)
cat("Correlation between conductivity and TDS:", cor_value, "\n")
```

```
Correlation between conductivity and TDS: NA
```

```
Code ▾
# Calculate R-squared (coefficient of determination)
r_squared <- cor_value^2
cat("R-squared:", r_squared, "\n")
```

```
R-squared: NA
```

```
Code ▾
# Fit linear model
lm_model <- lm(water_samples ~ water_samples)

# Get confidence intervals for the slope
confint(lm_model, level = 0.95)
```

```
Error in lm(water_samples ~ water_samples) : 
  contrasts can be applied only to factors with 2 or more levels
Run `rlang::last_error()` to see where the error occurred.
```

Create a linear regression model of the sample size with a least squares regression line and indications of slope and y intercept.

```
Code ▾
# Create linear regression model
lm_model_water <- lm(water_samples ~ water_samples)

# Summary of the model
summary(lm_model_water)
```

```
Error in lm(water_samples ~ water_samples) : 
  contrasts can be applied only to factors with 2 or more levels
Run `rlang::last_error()` to see where the error occurred.
```

```
Code ▾
# Create the plot with regression line
ggplot(plot_data, aes(x = conductivity, y = tds))
  + geom_point(alpha = 0.6, size = 2, color = "blue")
  + geom_smooth(method = "lm", se = TRUE, color = "red")
  + ggtitle("LSRL: Predicted y = 21374.6 + 1.5 (x)")
  + xlab("Conductivity of Water Bodies")
  + ylab("Total Dissolved Solids")
  + theme_minimal()
  + theme(plot.title = element_text
```

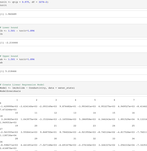
```
Error: unexpected '}' in "}"
Run `rlang::last_error()` to see where the error occurred.
```

Is there a statistically significant correlation to access of safe drinking water between Conductivity and TDS (ppm) across local, regional, and national levels?

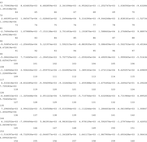
- The correlation between conductivity and TDS is significant across local, regional, and national levels.
- The p-value for each correlation test is extremely low (far smaller than 0.05), indicating a very high level of measurement error within the data.
- The correlation field remains a weak positive correlation, and shows no linearity.
- The LSRL indicates almost a perfectly horizontal slope. The slope is a positive one and lack of change over time.

```
Code ▾
# First, create a data frame combining all water quality data
plot_data_stats <- data.frame(conductivity = c(local_conductivity, regional_conductivity, national_conductivity),
```

densityx = water_statsConductivity)




```
Code ▾
qqnorm(plot_data_stats$conductivity)
```




```
Code ▾
plot(density(plot_data_stats$tds))
```

densityx = water_statsSolids)


```
Code ▾
qqnorm(plot_data_stats$tds)
```



```
Code ▾
# Calculate correlation between local and regional data
local_regional_cor <- cor(local_conductivity, regional_conductivity, use="complete.obs")
cat("Local-Regional correlation:", local_regional_cor, "\n")

# Calculate correlation between local and national data
local_national_cor <- cor(local_conductivity, national_conductivity, use="complete.obs")
cat("Local-National correlation:", local_national_cor, "\n")

# Calculate correlation between regional and national data
regional_national_cor <- cor(regional_conductivity, national_conductivity, use="complete.obs")
cat("Regional-National correlation:", regional_national_cor, "\n")
```

```
Local-Regional correlation: 0.0
Local-National correlation: 0.0
Regional-National correlation: 0.0
```

```
Code ▾
# Create a summary table
correlation_summary <- data.frac
```

```
Error: unexpected '<' in "correlation_summary <- data.frac"
Run `rlang::last_error()` to see where the error occurred.
```

```
[1,] -0.00752665 -0.00752665 -0.00752665 -0.00259907 -0.75459607 -1.15370904 -1.15370904 -1.15370904 -1.15370904 -1.15370904
[2,]  -0.00752665 -0.00752665 -0.00752665 -0.00259907 -0.75459607 -1.15370904 -1.15370904 -1.15370904 -1.15370904 -1.15370904
[3,]  -0.00752665 -0.00752665 -0.00752665 -0.00259907 -0.75459607 -1.15370904 -1.15370904 -1.15370904 -1.15370904 -1.15370904
```